TASKNEST · AI-USE NOTE

# AI did the heavy lifting. I made the calls and checked every number.

## 7 steps

From business question to a reusable skill

## 9 issues

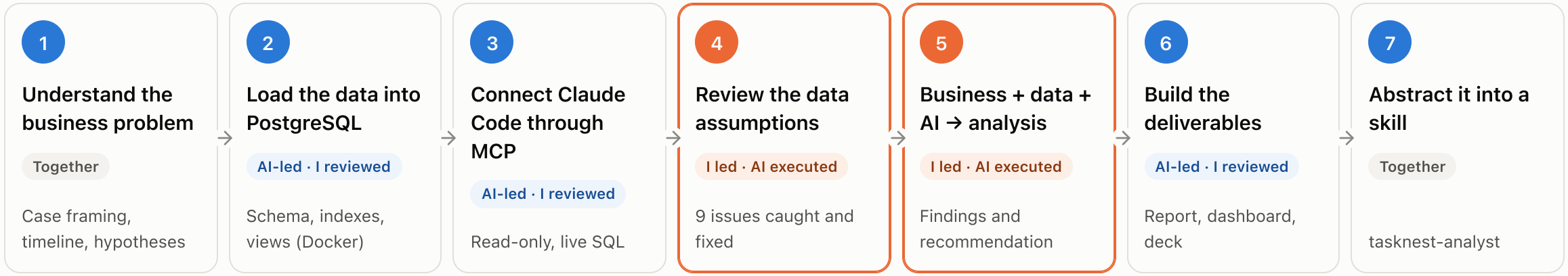
Caught in AI-drafted material before the CPO saw them

## 1 source

Every number is SQL on the raw tables

Tools: Claude Code (Claude) · PostgreSQL 16 in Docker · MCP server for Postgres · React + Vite · Playwright for automated checks

# The pipeline: from business question to a reusable skill




**The two steps that decide the answer (4 and 5) were led by me.** AI accelerated everything around them: setup, queries, statistics, code and drafts.

# Steps 2–3: a data layer built for scale and trust




## Scalable

The same SQL works on 1,800 rows or on production volumes.

## Reproducible

Every number is a query anyone can re-run.

## Safe

Read-only MCP: the AI can query the data, never modify it.

# Step 4: AI drafts are hypotheses. Nine issues caught before the CPO saw them

| What the draft said | What the data says | Type |
|---|---|---|
| ~~Revenue per trial Apr–May = $4.85~~ | $4.82 | Wrong number |
| ~~Price groups 434 / 970 / 396 users~~ | 450 / 986 / 364 | Wrong number |
| ~~“Guided wins in 100% of segments”~~ | True by channel and platform; 3 of 12 small cells negative | Overstated |
| ~~“Guided reduces price sensitivity”~~ | Price is just as inelastic in Classic before launch | Unsupported |
| ~~“Irrefutable, expand to 100% now”~~ | Observational evidence → expand to 90% with a holdout | Overconfident |
| ~~Activation view counts events from sign-up only~~ | Drops clock-skewed events: evening sign-ups −13 pp | Hidden bias |
| ~~Three “complete cohort” definitions in use~~ | Unified to one conservative cutoff | Inconsistency |
| ~~Pay-day histogram from a rounded view~~ | Recomputed unrounded (74% within 7 days holds) | Rounding |
| ~~Extraction time read from the MCP output~~ | MCP shifts timestamps by the UTC offset; fixed in SQL | Tooling |

# Step 5: the analysis was guided by me; AI executed it

**I decided**

- The real question: separate three overlapping changes, not just “did KPIs drop?”
- Fair comparison = same weeks (Guided vs Classic), never Pre vs Post
- Android 3.2 is a tracking problem, not a behavior change
- Method for the drop: rate/mix decomposition from my own BI analyst playbook
- Strength of the claim: observational → recommend a 90/10 holdout
- Go beyond the question: aha moment, price, timing, channels
- What to discard as artifact or noise

**AI executed**

- Wrote and ran dozens of SQL queries live through the MCP
- Computed confidence intervals, decompositions and stratified lifts
- Proposed hypotheses for me to test or reject
- Drafted the narrative, the dashboard code and the deck
- Re-ran checks every time a definition changed

**Accountability:** every number, assumption and recommendation was reviewed and is owned by me.

# Steps 6–7: one source of truth, four deliverables

**SQL on PostgreSQL**
raw tables · one definition per metric

↓

### Analysis report
Full storytelling and findings

- ✓ Every figure traced to a query; the rate/mix decomposition closes exactly to the total.

### Interactive dashboard
Timeline + segment comparisons

- ✓ SQL export feeds the app; automated browser tests matched SQL on every scenario.

### Decision deck
8 slides for the CPO

- ✓ Figures reconciled with the report and dashboard; every slide rendered and reviewed.

### tasknest-analyst skill
Lets stakeholders ask new questions

- ✓ 18 SQL templates executed and matched baselines; new-data playbook tested with a rolled-back load.

The skill encodes what was learned (definitions, pitfalls, methods, baselines), so the next analysis starts where this one ended.

# How I validated, and the limits

### Raw tables only
Recomputed from trial_users and product_events; pre-built views reviewed, not trusted

### Closure checks
Ratio decompositions sum exactly to the total change

### Reconciliation
Report = dashboard = deck = skill, number by number

### Artifact and noise tests
Every “insight” checked for data issues, mix and sample size

### Automated tests
18 SQL blocks executed; browser tests on the dashboard and deck

### Human review
I read and approved every claim and recommendation

**Limits**

- Synthetic case data
- Rollout not randomized → observational evidence; the holdout provides proof
- Revenue is first payment only (no LTV or churn yet)
- AI can be confidently wrong → treated as a draft until verified